

When a Control Does Not Control: A Mechanism-Derived Refutation of a False Positive in p -adic Cryptanalysis of the ECDLP

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Abstract

We report a systematic negative exploration of p -adic lifting attacks on the elliptic curve discrete logarithm problem (ECDLP) for ordinary curves with prime-order subgroups, of the kind underlying ECDSA. Across forty experimental phases every structural probe of the canonical-lift / formal-group attack surface came back inert. The methodological interest lies in one episode. A Gowers U^2 / Fourier analysis of the Teichmüller lift error $\delta(k)$ produced a persistent $5\text{--}11\sigma$ anomaly that a standard permutation control certified as “real, index-ordered structure.” We show that this conclusion was an artifact of a control that cannot distinguish the hypotheses it was meant to separate. Replacing it with two *mechanism-derived* nulls—one per candidate explanation—localizes the entire signal to a single exact, universal, public identity (the antisymmetry $\delta(k) + \delta(n - k) = [n]\tau(G)$), and shows it is neither new nor cryptanalytically useful. We then prove, as an explicit instance of a principle going back to Silverman and to Gadiyar–Padma, an inertness theorem that explains why *every* such anomaly on these curves must reduce to artifact-or-identity: the only secret-dependent part of any section lift error is governed by $k \bmod p^{e-1}$, which is information-theoretically independent of the public residue $k \bmod n$ whenever $\gcd(n, p) = 1$. The single inequality $\gcd(n, p) = 1$ is sharp: when it fails ($n = p$, anomalous curves) the obstruction vanishes and the same machinery recovers k with 100% success—Smart’s attack. The contribution is less the (essentially folklore) theorem than the discipline it certifies: in structure-hunting cryptanalysis, a control is valid only against a named alternative hypothesis, and nulls must be derived from mechanisms, not intuition.

1 Introduction

A recurring temptation in cryptanalysis is to lift a hard problem into a richer setting—a p -adic ring, a formal group, a higher-genus Jacobian—in the hope that structure invisible over $\mathbb{F}_p$ becomes visible and exploitable. For the ECDLP this hope is well mapped: Silverman’s “four faces of lifting” [1, 2] catalogues the local/global $\times$ torsion/non-torsion strategies and the obstruction each meets, and the only p -adic local lift that ever broke an ECDLP is Smart’s attack on anomalous curves [4, 5, 6], where the group order equals the characteristic.

This paper is a case study in *how one knows when a lifting probe has actually found nothing*. We conducted a long, deliberately exhaustive negative exploration of the canonical-lift attack surface for ordinary curves with large prime-order subgroups—the ECDSA regime—and every phase was inert. Such a project invites a methodological hazard: after dozens of null results, a probe eventually

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returns an “anomaly,” and the analyst must decide whether it is signal. We hit exactly this case, initially mis-called it signal on the strength of a textbook control, and only a more careful experiment overturned it. We think the overturning, and the reason the anomaly *had* to be spurious, are more useful to report than the negative result itself.

Contributions.

- A worked, quantitative example of a *control that does not control*: a permutation test that appears to certify structure but provably cannot separate the competing explanations (§4).
- A general recipe—*mechanism-derived controls*: enumerate the candidate generating mechanisms and build one null per mechanism—and its application, which localizes a 5–11 σ Gowers anomaly to a single exact public identity (§5).
- An inertness theorem (§6) that explains why such anomalies are forced: for $\gcd(n, p) = 1$, the secret-dependent part of any section lift error lives in a coordinate $(k \bmod p^{e-1})$ independent of the public one $(k \bmod n)$. We state it as the explicit instance it is of [1, 3], with full proof and numerical verification.
- A sharp-boundary demonstration (§7): the same construction, run on an anomalous curve where $\gcd(n, p) = p$, recovers the secret with 100% success. The theorem’s one hypothesis is exactly the line between inert and broken.

We are explicit throughout that the theorem is not new mathematics; its role is to certify, after the fact, that the methodological suspicion was structurally warranted.

2 Background and notation

Let $p \geq 5$ be prime and $E/\mathbb{F}_p$ an elliptic curve with $\#E(\mathbb{F}_p) = N$. Fix $G \in E(\mathbb{F}_p)$ of order n with $\gcd(n, p) = 1$, and $Q = kG$; the ECDLP is to recover $k \in \mathbb{Z}/n\mathbb{Z}$ from (E, G, Q) . For $e \geq 1$, reduction modulo p on a fixed Weierstrass lift gives the exact sequence of abelian groups

$$0 \longrightarrow \widehat{E}(p\mathbb{Z}/p^e) \longrightarrow E(\mathbb{Z}/p^e) \xrightarrow{\text{red}} E(\mathbb{F}_p) \longrightarrow 0, \quad (1)$$

where the kernel of reduction $\widehat{E}(p\mathbb{Z}/p^e)$ is a p -group of order p^{e-1} . The formal parameter $z = -X/Y$ identifies $\widehat{E}$ with $(p\mathbb{Z}/p^e, +_F)$ ($+_F$ the formal group law), and the formal logarithm $\log: \widehat{E} \rightarrow (p\mathbb{Z}/p^e, +)$ is an isomorphism with $\log([m]u) = m \log(u)$. For $E: y^2 = x^3 + b$ the log has the sparse form $\log(z) = z + \frac{b}{7}z^7 + \dots$, so for $z \in p\mathbb{Z}$ and $e \leq 4$ one has $\log(z) \equiv z \pmod{p^e}$: kernel arithmetic is additive in z at our precision.

A *section* is a map $\tau: E(\mathbb{F}_p) \rightarrow E(\mathbb{Z}/p^e)$ with $\text{red} \circ \tau = \text{id}$; the Teichmüller and Hensel lifts are sections, generally not homomorphisms. The *lift error* (a 1-cochain valued in $\widehat{E}$ by (1)) is

$$\delta_\tau(k) := [k] \tau(G) - \tau(kG) \in \widehat{E}(p\mathbb{Z}/p^e). \quad (2)$$

We write $\delta(k)$ for the Teichmüller case and $z(\delta(k)) \in p\mathbb{Z}/p^e$ for its formal parameter.

Prior art. That a p -adic lift transports the ECDLP without weakening it, that the canonical (Teichmüller) lift carries no information, and that “solving the DLP is equivalent to constructing a non-canonical lift” are due to Silverman [1, 2] for elliptic curves and to Gadiyar–Maini–Padma and Gadiyar–Padma [3] for $\mathbb{F}_p^\times$. Smart, Satoh–Araki and Semaev [4, 5, 6] give the anomalous attack. The canonical lift itself is Lubin–Serre–Tate [7]; its computational instance is Satoh’s point-counting algorithm [8]. Our theorem is an explicit packaging of these in the lift-error language.

3 The exploration, in brief

The full study ran forty numbered phases against the canonical-lift / formal-group attack surface for $j = 0$ (CM by $\mathbb{Z}[\omega]$) and generic ordinary curves. We summarize only the shape of the result; details are in the repository. Probes included: injectivity and pseudorandomness of δ (DFT flatness, maximal Berlekamp–Massey complexity, uniform p -adic valuation, no polynomial/Mahler/multiplicative fit); a homomorphic section from H^2 vanishing (which exists but, being a homomorphism, annihilates the attack); an explicit Serre–Tate canonical lift validated by $\delta_{\text{can}} \equiv 0$; HNP/lattice (LLL, BKZ) attacks on the cocycle; MOV embedding degree; ordinary/supersingular bridges; genus-2 Jacobian covers; Weil pairings; Iwasawa μ ; and a Frobenius/unit-root analysis. Every one was inert. Two exact relations survived, both *public* (computable from the curve) and both giving no advantage: the automorphism-equivariance rank law $\text{rank} = \min(\cdot, (n-1)/|\text{Aut}(E)|)$ for the leading digit of δ , and the antisymmetry

$$\delta(k) + \delta(n-k) = [n] \tau(G), \quad (3)$$

verified to hold *exactly* in the z -coordinate (e.g. for $p = 97$, $z(\delta(k)) + z(\delta(n-k))$ takes a single value across all 78 pairs).

4 A false positive: the Gowers anomaly

Phases of the study applied higher-order Fourier analysis to the digit sequence of δ . For digit j encode $f_j(k) = \exp(2\pi i \text{digit}_j(z(\delta(k)))/p)$ and measure the Gowers U^2 norm against an S^1 -random baseline. Across $p = 31$ to 521 the U^2 norm was elevated $5\text{--}11\sigma$; U^3 and higher were at baseline. The leading Fourier coefficient decayed as $|\hat{f}(\alpha)| \sim Cn^{-\beta}$ with $\beta \in [0.28, 0.47] < 1/2$, present on CM, $j=1728$, and non-CM curves alike. An internal report concluded this was a genuine, previously unnoticed degree-1 structural property of the Teichmüller cocycle “not explained by any universal identity.”

The evidence offered was a permutation control: randomly permuting the *values* of f_j destroyed the elevation, which was read as proof that the structure lived in the index ordering, hence was “real.” This inference is invalid, and seeing why is the point of the paper.

5 Mechanism-derived controls

The principle. A control is a null model, and a null model tests exactly one alternative hypothesis: the one under which the statistic is recomputed. The value-permutation control recomputes U^2 under “the multiset of digit values, in random order.” It therefore separates “structure in the value distribution” from “structure in the ordering”—and nothing else. But at least three distinct mechanisms predict ordered structure here, and the permutation kills all of them simultaneously, so it cannot attribute the signal to any one:

- (H1) *Prefix-sum artifact.* $\delta(k) = -\sum_{i < k} c(iG, G)$ is a running sum. Summation is a low-pass filter ($\hat{s}(\xi) = \hat{d}(\xi)/(1 - e^{2\pi i \xi/n})$, a $1/\xi$ pole at $\xi = 0$), so *any* increments with this multiset, once summed, carry low-frequency mass and elevated U^2 .
- (H2) *Antisymmetry shadow.* The exact reflection (3) couples Fourier mode α with $-\alpha$, a rigid constraint that elevates U^2 .
- (H3) *Genuine new structure,* the claimed explanation.

p	L	$z(\text{real vs incShuf})$	$z(\text{incShuf vs rand})$
67	78	+3.8 to +6.4	−0.24 to +0.05
97	78	+3.6 to +6.2	+0.17 to +0.48
167	166	+6.6 to +8.2	+0.09 to +0.67
257	256	+6.7 to +8.4	−0.11 to +0.99
521	520	+10.3 to +11.9	−0.07 to +0.35

Table 1: Phase 40 increment-shuffle control (digits $j = 1, 2, 3$). The shuffled-increment prefix sums match the random baseline (right column ≈ 0); the true ordering is far above and grows with n (middle). H1 (prefix-sum artifact) is refuted. The half-sequence control (Phase 40b) then collapses the middle column to noise, localizing the signal to the antisymmetry (3).

The fix is to build one null per mechanism, each derived from how that mechanism generates ordering, and to ask which null the real statistic matches.

Control for H1 (increment shuffle). $s(k) = z(\delta(k))$ is the exact cumulative sum of its first differences $d(k) = s(k+1) - s(k)$. Shuffle d , rebuild s' by re-summing in the new order, recompute U^2 ; the increment multiset is identical, only the order changes. Table 1 (Phase 40). For all primes with sequence length $L \geq 66$ the shuffled-increment prefix sums sit *at* the random baseline ($z(\text{incShuf vs rand})$ within ± 1 , range -0.26 to $+1.04$), while the true ordering sits $+4$ to $+12$ above them, growing with n ; the two smallest primes ($L \leq 30$) are dominated by baseline noise and excluded. So H1 is refuted: the digit nonlinearity destroys the random walk’s low-frequency mass, and the signal is in the *specific* order, not in summation.

Control for H2 (half-sequence). The reflection (3) relates each k to $n - k$. Restricting to the first half $k = 1, \dots, \lfloor (n-1)/2 \rfloor$ removes every reflection pair from the window, so H2 predicts no internal constraint there; H3 (an intrinsic ordering structure) predicts the signal persists. Re-running the Phase 40 control on the half (Phase 40b): the elevation *vanishes*, $z(\text{real vs incShuf})$ scattering in $[-2.4, +2.1]$ with no growth in n . At matched sample size the contrast is decisive: $p = 257$ gives $z \approx +8$ on the full sequence ($L = 256$) and $z \approx +0.1$ to $+0.6$ on the half ($L = 128$).

Verdict. The anomaly is the Fourier-domain shadow of the exact, universal, public identity (3): H2, not H1 or H3. It is real but not new, and—being a known public consequence of the curve’s geometry that yields only a 2-to-1 search reduction—carries no cryptanalytic advantage. The varying “decay exponent” $\beta \in [0.28, 0.47]$ and its apparent curve-family dependence are finite-size noise around a single reflection-induced spectrum; nothing family-specific survives the half-sequence control.

6 Why it had to be a false positive: an inertness theorem

The episode above was not luck. For $\gcd(n, p) = 1$ *every* secret-dependent feature of a section lift error is forced into a coordinate the public data cannot see, so any apparent “structure” in δ that is computable from (E, G, Q) must reduce to a public identity or an artifact. We make this precise.

Lemma 1 (Canonical lift). *If $\gcd(n, p) = 1$, the order- n subgroup $\langle G \rangle$ lifts uniquely to $\langle \tilde{G} \rangle \subset E(\mathbb{Z}/p^e)$ of order n with $\text{red}(\tilde{G}) = G$, and $jG \mapsto [j]\tilde{G}$ is a homomorphism (so $\delta_{\tau_{\text{can}}} \equiv 0$). Explicitly $\tilde{G} = [m]\tau(G)$ for any section τ , where $m \equiv 1 \pmod{n}$, $m \equiv 0 \pmod{p^{e-1}}$.*

Proof. $E(\mathbb{Z}/p^e)[n]$ injects into $E(\mathbb{F}_p)[n]$ (an n -torsion element of $\widehat{E}$ is killed by $\gcd(n, p^{e-1}) = 1$) and surjects onto it (étale n -torsion lifts by Hensel), so reduction is an isomorphism on n -torsion and $\widetilde{G}$ is the unique order- n lift of G . Writing $\tau(G) = \widetilde{G} + c$ with $c \in \widehat{E}$, $[m]\tau(G) = [m \bmod n]\widetilde{G} + [m \bmod p^{e-1}]c = \widetilde{G}$. $\square$

Theorem 1 (Inertness). *Let τ be any section and $c := \tau(G) - \widetilde{G} \in \widehat{E}$. Then:*

- (a) $\delta_\tau(k) = [k]c - c_k$, where $c_k := \tau(kG) - [k]\widetilde{G} \in \widehat{E}$ depends only on the point kG .
- (b) $[k]c$ depends on k only through $k \bmod p^{e-1}$; $\log([k]c) = k \log(c)$ has additive order a power of p dividing p^{e-1} .
- (c) Since $\gcd(n, p^{e-1}) = 1$, the residues $k \bmod n$ (fixed by Q) and $k \bmod p^{e-1}$ (governing $[k]c$) are independent coordinates of k ; no value of $[k]c$ is determined by (E, G, Q) .
- (d) δ_τ is a well-defined function of the instance (i.e. of $k \bmod n$) iff $c = 0$ (i.e. τ canonical on $\langle G \rangle$), in which case $\delta_\tau \equiv 0$. For $c \neq 0$, $\delta_\tau(k+n) - \delta_\tau(k) = [n]c = [n]\tau(G) \neq O$, the exact antisymmetry constant of (3).

Proof. (a) $[k](\widetilde{G} + c) - ([k]\widetilde{G} + c_k) = [k]c - c_k$. (b) $c \in \widehat{E}$ has order dividing p^{e-1} ; apply $\log$. (c) CRT. (d) c_k is a function of kG , hence of $k \bmod n$; thus δ_τ factors through $k \bmod n$ iff $k \mapsto [k]c$ does, iff $[n]c = O$, iff (as $\gcd(n, p^{e-1}) = 1$) $c = O$. The shift equals $[n]c = [n]\tau(G)$ because $[n]\widetilde{G} = O$. $\square$

Corollary 1. *For $\gcd(n, p) = 1$ no section lift error is a function of (E, G, Q) that depends on the secret: it is either identically zero (canonical) or multivalued with ambiguity $[n]\tau(G) \neq O$ (non-canonical). The p -adic canonical lift moves the ECDLP without weakening it.*

This is exactly why the Gowers anomaly was forced to be artifact-or-identity. The only secret-bearing object, $[k]c$, is invisible to the attacker; the visible δ is either zero or, taken as a function of the integer k , periodic-with-shift $[n]\tau(G)$ —and that shift is the reflection (3) whose Fourier signature Phase 37 rediscovered.

7 The boundary is sharp

Theorem 1 hinges on the single inequality $\gcd(n, p) = 1$. Set $n = p$ (an anomalous curve, $\#E(\mathbb{F}_p) = p$) and every clause degenerates toward the attacker: $k \bmod n = k \bmod p$ now coincides with the residue the formal group sees, and the obstruction $[n]c = [p]c$ vanishes because $c \in \widehat{E}(p\mathbb{Z}/p^2)$ has order dividing p . The lift error becomes well-defined and the formal logarithm of $[p] \cdot (\text{lift})$ is a homomorphism $E(\mathbb{F}_p) \rightarrow \mathbb{F}_p$ recovering k —Smart’s attack.

We verified both sides numerically (Table 2). On every ordinary test curve the obstruction $[n]\tau(G)$ is a nonzero kernel element of *maximal* order p^{e-1} (valuation 1); on an anomalous curve $y^2 = x^3 + 4x + 7$ over $\mathbb{F}_{53}$ the obstruction vanishes and Smart’s attack recovers k with 100% success (12/12 random secrets). The same code path that is inert for $\gcd(n, p) = 1$ becomes a total break the instant $n = p$.

8 Numerical verification

Theorem 1 was checked on the secp256k1 family $y^2 = x^3 + 7$ at $e = 4$ for ten primes $p = 31, \dots, 163$ with $\gcd(n, p) = 1$ (Phase 41). All kernel arithmetic was done through the additive z -coordinate to avoid projective degeneracy on deep kernel points. On every prime: the obstruction $C_n :=$

Regime	curves	$\gcd(n, p)$	obstruction $[n]\tau(G)$	ECDLP
Ordinary (ECDSA)	$p = 31\text{--}163, j=0$	1	nonzero, $\text{ord} = p^{e-1}$	inert
Anomalous (Smart)	$y^2 = x^3 + 4x + 7/\mathbb{F}_{53}$	p	O (vanishes)	broken, 12/12

Table 2: Phases 41–42. The well-definedness obstruction of Theorem 1(d) is nonzero of maximal order on ordinary curves and vanishes on anomalous curves, where Smart’s attack succeeds. The hypothesis $\gcd(n, p) = 1$ is the exact boundary.

$z([n]\tau(G))$ is nonzero and lies in $p\mathbb{Z}$ with valuation 1, of additive order exactly p^{e-1} (clause d, b); the antisymmetry $z(\delta(k)) + z(\delta(n-k)) \equiv C_n$ holds for all k (clause d / (3)); the integer-representative shift $z(\delta(k+n)) - z(\delta(k)) \equiv C_n$ holds on all eight arithmetically-reliable primes (clause d; it is a group identity, and on the two remaining primes the deep multiply degenerates in exact arithmetic, flagged by a self-consistency probe); and $\gcd(n, p^{e-1}) = 1$ (clause c). The canonical-lift case $\delta_{\text{can}} \equiv 0$ (Lemma 1) was confirmed independently in an exact p -adic computation. Anomalous recovery is Phase 42 (§7).

9 Discussion

For negative cryptanalysis. The reusable lesson is narrow and practical. (i) A control tests exactly one alternative—the one its null recomputes. Before trusting it, enumerate the mechanisms that could generate the observed statistic and check whether the control separates them; the permutation test here failed precisely because three mechanisms collapsed under it. (ii) Build one null per mechanism, *derived from how that mechanism generates the statistic*: H1 from the prefix-sum structure (shuffle increments, re-sum), H2 from the reflection’s reach (restrict to a half). (iii) Beware cumulative statistics: any running sum manufactures low-frequency autocorrelation, so “looks autocorrelated” is the null, not the finding. (iv) Where a structural theorem is available, it bounds what controls can ever find; Theorem 1 says no honest control will turn δ into an oracle for $\gcd(n, p) = 1$.

For automated and AI-assisted exploration. Pipelines that fan out many structural probes and surface the “most anomalous” are increasingly common, and they are false-positive engines: with enough encodings and transforms, an apparent $5\text{--}11\sigma$ effect is cheap. The safeguard is not more probes but mechanism-derived nulls and, where possible, a closed-form reason the probe must be inert. The value of the theorem in this study is exactly that: it converts “we tried forty things and they failed” into “here is the one-line reason they had to.”

Honest scope. Theorem 1 is not new mathematics. It is an explicit instance, in the lift-error language, of the principle of [1, 2, 3] that lifting transports the ECDLP; the anomalous degeneration is Smart’s attack [4]. It concerns section lift errors only; pairing-based functionals are a separate mechanism (inert here for the independent reason of huge embedding degree). We claim novelty only for the methodology and for the explicit identification of the obstruction with the antisymmetry constant.

10 Conclusion

A long negative exploration of p -adic ECDLP lifting produced one seductive anomaly. A standard control endorsed it; mechanism-derived controls killed it, attributing it entirely to an exact public identity; and an inertness theorem explains why no such anomaly on ordinary curves can ever be more than artifact-or-identity, with the boundary falling exactly at the anomalous curves where the lift attack genuinely works. The cryptanalytic verdict is the expected negative one. The transferable result is a discipline: name your alternative hypotheses, build a null for each from its mechanism, and prefer a structural reason for inertness over an accumulation of passed tests.

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